THE TAYLOR COEFFICIENTS OF THE RIEMANN ξ-FUNCTION FORM A PÓLYA FREQUENCY SEQUENCE OF ORDER 3

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Abstract. Let $\xi(s) = \frac{1}{2}s(s-1)\pi^{-s/2}\Gamma(s/2)\zeta(s)$ and write $\xi(\frac{1}{2}+z) = \sum_{n\geq 0}\gamma_n z^{2n}/n!$, so that all $\gamma_n > 0$. We prove, unconditionally, that the normalized sequence $a_n = \gamma_n/n!$ is a Pólya frequency sequence of order 3: every minor of order at most 3 of the one-sided Toeplitz matrix $(a_{j-i})_{i,j\geq 0}$ is nonnegative. By the theory of Aissen–Schoenberg–Whitney–Edrei, the corresponding statement for minors of *every* order is equivalent to the Riemann Hypothesis; order 2 (the Turán inequalities) was established by Csordas, Norfolk and Varga in 1986, and the degree-3 statement on the parallel Jensen-polynomial ladder by Dimitrov and Lucas in 2011.

The proof combines four ingredients: (i) a structure theorem reducing all consecutive-row order-3 minors of a positive one-sided Toeplitz sequence to four ratio conditions; (ii) a completion theorem, proved by two signed Grassmann–Plücker inductions, showing that for strictly log-concave sequences the consecutive-row case already implies full PF_3; (iii) certified interval-arithmetic verification of the ratio conditions for $2 \leq n \leq 126$; and (iv) an analytic proof of the ratio conditions for all $n \geq 127$ via two-sided bounds on the cumulants of an exponentially tilted moment family, with explicit rational constants.

The result is sharp for the method: we exhibit explicit rational sequences satisfying all four ratio conditions, and even full window-PF_3, for which an order-4 Toeplitz minor is negative. Consequently no ratio-type hypothesis of the kind used here can reach PF_4, and further progress on the Pólya frequency ladder for ξ requires structural information specific to ξ. All computational claims are backed by a reproducible artifact with fail-closed manifests, pinned dependency hashes, and deterministic replay.

1. Introduction

Write

$$\xi(s) = \tfrac{1}{2}\, s(s-1)\,\pi^{-s/2}\,\Gamma(s/2)\,\zeta(s), \qquad \Psi(z) := \xi(\tfrac{1}{2}+z) = \sum_{n\geq 0}\frac{\gamma_n}{n!}\,z^{2n}, \tag{1}$$

so that $\gamma_n > 0$ for all $n \geq 0$. The Riemann Hypothesis (RH) asserts that all zeros of Ψ are purely imaginary; equivalently, setting

$$F(w) := \Psi(\sqrt{w}) = \sum_{n\geq 0} a_n w^n, \qquad a_n := \frac{\gamma_n}{n!} > 0, \tag{2}$$

RH asserts that the entire function F, which has genus 0, has only real nonpositive zeros.

A classical theorem of Aissen, Schoenberg and Whitney, completed by Edrei [1, 2], characterizes exactly this situation in terms of total positivity: a sequence $(a_n)_{n\geq 0}$ of nonnegative numbers is a *Pólya frequency sequence* (PF_∞), i.e. every finite minor of the one-sided Toeplitz matrix

$$T(a) = (a_{j-i})_{i,j\geq 0}, \qquad a_k := 0 \text{ for } k < 0,$$

is nonnegative, if and only if its generating function has the form $Cz^m e^{\gamma z}\prod_i(1+\alpha_i z)/(1-\beta_i z)$ with $C, \gamma, \alpha_i, \beta_i \geq 0$. For the genus-0 function F of (2) this reduces to: (a_n) *is* PF_∞ *if and*

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only if RH holds. Total positivity of finite order therefore provides a ladder of unconditional statements converging to RH: for $r \geq 1$, the sequence is PF_r when every minor of $T(a)$ of order at most r is nonnegative. Classical sources and modern treatments of this circle of ideas include [9, 12, 10, 11, 8].

The history of this ladder, and of its close relative built from Jensen polynomials, is short. Order 2 — log-concavity of (a_n), which amounts to $\gamma_n^2 \geq \frac{n}{n+1}\gamma_{n-1}\gamma_{n+1}$ and is therefore implied by (and slightly weaker than) the Turán inequalities $\gamma_n^2 \geq \gamma_{n-1}\gamma_{n+1}$ — follows from the work of Csordas, Norfolk and Varga [3]. On the Jensen ladder, hyperbolicity of the degree-3 Jensen polynomials for all shifts (the "higher order Turán inequalities") was proved by Dimitrov and Lucas [4]. Griffin, Ono, Rolen and Zagier [5] (see also [6, 7]) later proved that for each fixed degree d the Jensen polynomials $J^{d,n}$ are hyperbolic for all sufficiently large shifts n. Families of *contiguous* order-3 determinants attached to ξ also appear in the determinantal programme of Nuttall [14], which discusses positivity for particular contiguous subfamilies. To our knowledge, however, no unconditional statement covering *all* order-3 Toeplitz minors of (a_n) — sparse rows and columns included — has appeared. The present paper closes this order:

Theorem 1.1 (Main Theorem)**.** *The sequence $a_n = \gamma_n/n!$ of normalized Taylor coefficients of ξ is PF_3: every minor of order at most 3 of the one-sided Toeplitz matrix $T(a)$ is nonnegative. Moreover the order-1 entries and all nonzero order-2 minors are strictly positive, and the contiguous order-3 minors are strictly positive.*

The proof is organized around two abstract theorems and two ξ-specific inputs.

- **(A)** A *structure theorem* (Theorem 2.2): for any positive one-sided Toeplitz sequence, four ratio conditions — log-concavity, monotonicity of the second-order ratios q_n, the initial bound $q_2 \leq \frac{1}{2}$, and nonnegativity of the *contiguous* order-3 minors — imply nonnegativity of every order-3 minor with three *consecutive rows* (arbitrary columns).
- **(B)** A *completion theorem* (Theorem 2.3): for strictly log-concave positive sequences, the consecutive-row case of order 3 already implies full PF_3. The proof runs two signed three-term Grassmann–Plücker inductions in which every divisor is strictly positive and no sign-unknown minor ever appears with a negative coefficient.
- **(C)** *Certified finite verification*: the four ratio conditions hold for $2 \leq n \leq 126$, verified in ball arithmetic (Arb [18]) with rigorous enclosures of $\gamma_2, \ldots, \gamma_{130}$ and directed rounding throughout.
- **(D)** An *analytic tail*: for all $n \geq 127$ the ratio conditions follow from new two-sided bounds on the second and third cumulants of an exponentially tilted moment family attached to the classical integral representation of Ξ, with explicit rational constants; the conversion to the discrete conditions is a finite list of coefficient-positive polynomial identities.

The result is *sharp for its method* in a strong sense.

Theorem 1.2 (Sharpness; see §6)**.** *There exist explicit rational positive sequences satisfying all four ratio conditions of (A) — indeed satisfying every order-3 nonnegativity requirement on a window containing the failure — for which an explicit order-4 Toeplitz minor is negative. There also exists a positive sequence with entire generating function satisfying the same ratio conditions for which already the degree-3 Jensen polynomial $J^{3,0}$ fails to be hyperbolic.*

Thus ratio-type conditions of the kind that power Theorem 1.1 cannot prove PF_4, nor even degree-3 Jensen hyperbolicity, for any abstract sequence: any ascent of the ladder beyond order 3 must use information specific to ξ that is invisible to these invariants. We regard this negative statement as of equal practical importance to the positive one; it delimits precisely where the present technique stops. Under RH, of course, (a_n) is PF_r for every r; our countermodels are abstract sequences, not related to ξ.

Relation to the Jensen ladder. Theorem 1.1 and the Dimitrov–Lucas theorem [4] are statements on two different ladders which intersect only at infinity. Hyperbolicity of degree-d Jensen polynomials is a discriminant-type condition; PF_r is a determinantal one. At finite level neither family of statements implies the other in any obvious way, and the countermodels of §6 separate them explicitly in one direction. The two ladders coincide, by [1, 2] and by Pólya's classical equivalence, only in the limit $r = \infty/d = \infty$, where both are RH.

Computer assistance and reproducibility. Steps (C) and parts of (D) are computer-assisted in the tradition of [17]: every numerical claim entering the proof is a certified ball-arithmetic enclosure with outward rounding, produced with Arb through the open-source `python-flint` bindings and assembled by a fail-closed checker that pins the exact inputs by cryptographic hash and refuses to report success unless every dependency replays. The complete artifact (code, certificates, manifests, replay instructions) accompanies the paper; see §7. The research programme that produced the mathematics was conducted by autonomous AI reasoning agents under human direction and audit; we describe the workflow and the verification discipline in §9, and emphasize that the proof as presented is human-readable and independent of any claim about how it was found.

2. Notation and main results

Throughout, $(a_n)_{n\geq 0}$ is a sequence with $a_n > 0$, extended by $a_k = 0$ for $k < 0$. Define

$$R_n = \frac{a_n}{a_{n-1}} \ (n \geq 1), \qquad q_n = \frac{R_n}{R_{n-1}} \ (n \geq 2). \tag{3}$$

Log-concavity of (a_n) (PF_2) is $q_n \leq 1$ for all $n \geq 2$; *strict* PF_2 is $q_n < 1$, i.e. R_n strictly decreasing.

For increasing index triples $R = (r_0, r_1, r_2)$, $C = (c_0, c_1, c_2)$ set

$$\Delta_R^C = \det\big(a_{c_j - r_i}\big)_{0\leq i,j\leq 2}.$$

A minor is *contiguous* when both R and C are triples of consecutive integers; it has *consecutive rows* when R is. For $m \geq 0$ we write

$$D_m := \Delta_{(0,1,2)}^{(m,m+1,m+2)}$$

for the *translated contiguous determinant at start m*; by the Toeplitz structure every contiguous minor is some D_m.

Definition 2.1. We say (a_n) satisfies the *ratio conditions* when:

(R1) $q_n \leq 1$ for all $n \geq 2$ (PF_2);
(R2) $q_{n+1} \geq q_n$ for all $n \geq 2$;
(R3) $q_2 \leq \frac{1}{2}$;
(R4) every contiguous order-3 minor is nonnegative.

Theorem 2.2 (Consecutive rows from ratio conditions)**.** *If (a_n) satisfies the ratio conditions, then every order-3 minor of $T(a)$ with consecutive rows is nonnegative.*

Theorem 2.3 (Plücker completion)**.** *Let (a_n) be positive and strictly PF_2, and suppose every order-3 minor of $T(a)$ with consecutive rows is nonnegative. Then (a_n) is PF_3.*

Theorem 2.4 (Analytic and certified inputs for ξ)**.** *For $a_n = \gamma_n/n!$:*

(i) *$a_n > 0$ for all $n \geq 0$;*
(ii) *$q_n < 1$ for all $n \geq 2$ (strict PF_2);*
(iii) *$q_2 < \frac{1}{2}$;*
(iv) *$q_{n+1} \geq q_n$ for all $n \geq 2$;*
(v) *every contiguous order-3 minor is strictly positive.*

Theorem 1.1 follows at once: Theorem 2.4 verifies the ratio conditions for ξ, Theorem 2.2 yields all consecutive-row order-3 minors, and Theorem 2.3 completes to PF_3.

3. Consecutive rows from the ratio conditions

Fix rows $(m, m+1, m+2)$; by the Toeplitz structure it suffices to treat $m = 0$, replacing columns (c_0, c_1, c_2) by their translates. Columns containing an index < 0 produce structurally triangular or zero minors: if $c_0 < 0$ the first column has at most one nonzero entry and the minor reduces, with nonnegative cofactor by PF_2, to an order-2 case. So assume $0 \leq c_0 < c_1 < c_2$ and write the columns as (c_0, c_1, c_2).

The proof splits into three families according to c_0.

3.1. **Columns** $(0, j, k)$**.**

An exact factorization gives

$$\Delta^{(0,j,k)}_{(0,1,2)} = a_0\, a_{j-2}\, a_{k-2}\,\bigl(R_{j-1} - R_{k-1}\bigr), \qquad 2 \leq j < k, \tag{4}$$

which is nonnegative by (R1) since $j - 1 < k - 1$. (For $j \leq 2$ the minor is triangular.)

3.2. **Columns** $(1, j, k)$**.**

With $G_c := R_{c-1}\,(R_1 - R_c)$ one has the exact factorization

$$\Delta^{(1,j,k)}_{(0,1,2)} = a_0\, a_{j-2}\, a_{k-2}\,\bigl(G_j - G_k\bigr), \qquad 2 \leq j < k. \tag{5}$$

Monotonicity of $c \mapsto G_c$ on $c \geq 2$ is exactly where (R2) and (R3) enter: writing $G_{c+1} - G_c$ in terms of the q's, the anchored inequality $q_2 \leq \frac{1}{2}$ together with q nondecreasing forces G nonincreasing, hence (5) is nonnegative.

3.3. **Columns** (i, j, k)**,** $i \geq 2$**.**

Here the minor factors as a positive product times the orientation determinant of the three planar points

$$P_c = \bigl(R_{c-1},\, R_{c-1}R_c\bigr), \qquad c \in \{i, j, k\},$$

taken in decreasing abscissa order (by strict monotonicity of R). Nonnegativity of all such orientations is a discrete convexity statement for the polygonal curve $c \mapsto P_c$; the contiguous case $(i, i+1, i+2)$ is precisely condition (R4), and the general case follows from the contiguous one because convexity of consecutive triples of a planar polygonal chain with monotone abscissae implies convexity of all triples.

A complete symbolic expansion of the three families, including all boundary and triangular cases, is replayed in the artifact (§7); the identities (4)–(5) are elementary to verify by direct expansion. □

Remark 3.1. Conditions (R1)–(R4) are close to minimal for Theorem 2.2: dropping (R3), or replacing (R2) by any bounded number of its instances, admits counterexamples already at order 3; see the artifact for exact witnesses. The sharpness examples of §6 show the conclusion cannot be strengthened to PF_4.

4. The Plücker completion

We prove Theorem 2.3. All identities keep the column triple $C = (c_0, c_1, c_2)$ fixed and vary rows; $\Delta_{(x,y,z)}$ abbreviates $\Delta^{C}_{(x,y,z)}$.

4.1. **Strict order-2 positivity for arbitrary gaps.** For rows $i < j$ and columns $p < q$ the order-2 minor is $a_{p-i}a_{q-j} - a_{q-i}a_{p-j}$. If $i \le p < j \le q$ it is triangular with strictly positive diagonal. In the interior case $p \ge j$, put $x = p - j$, $\alpha = j - i$, $\beta = q - p$; then

$$a_{x+\alpha}a_{x+\beta} - a_{x+\alpha+\beta}a_x = a_x a_{x+\beta}\Big(\prod_{t=1}^{\alpha} R_{x+t} \;-\; \prod_{t=1}^{\alpha} R_{x+\beta+t}\Big) > 0, \tag{6}$$

strictly, term by term, because R is strictly decreasing. Thus every order-2 minor is nonnegative, and strictly positive unless it is structurally zero.

4.2. **The auxiliary row $d = c_1 + 1$.** For any rows $i < j < d$ with $d := c_1 + 1$, the row of $T(a)$ indexed by d restricted to the columns C is $(0, 0, a_{c_2-c_1-1})$, whence the exact factorization

$$\Delta_{(i,j,d)} = a_{c_2-c_1-1}\ \det\begin{pmatrix} a_{c_0-i} & a_{c_1-i} \\ a_{c_0-j} & a_{c_1-j}\end{pmatrix}, \tag{7}$$

whose scalar prefix is strictly positive (equal to a_0 at the edge $c_2 = d$) and whose determinant is an order-2 minor, nonnegative by (6).

4.3. **Boundary reductions.** Three one-sided cases are removed before any division: if $c_2 < r_2$ the last row vanishes; if $c_1 < r_2 \le c_2$, expansion along the last row factors the minor as a positive entry times an order-2 minor; if $c_0 < r_1 \le c_1$, expansion along the first column does the same. The remaining *inductive region* is

$$c_1 \ge r_2 \quad \text{and} \quad c_0 \ge r_1, \tag{8}$$

in which $d = c_1 + 1$ exceeds every row in play and all divisors below are strictly positive interior order-2 cases of (6) (or the triangular edge case).

4.4. **Adjacent-pair rows.** Let $R = (r, r+1, s)$ with $s > r+2$. The three-term Grassmann–Plücker identity on the five rows $r, r+1, r+2, s, d$ reads

$$\Delta_{(r,r+1,s)}\ \Delta_{(r+1,r+2,d)} = \Delta_{(r,r+1,r+2)}\ \Delta_{(r+1,s,d)} + \Delta_{(r,r+1,d)}\ \Delta_{(r+1,r+2,s)}. \tag{9}$$

On the right, the first factor has consecutive rows (nonnegative by hypothesis), the two d-minors are nonnegative by (7), and the last factor has the same adjacent pair with row gap smaller by one; the divisor $\Delta_{(r+1,r+2,d)}$ is strictly positive in the region (8). Induction from the consecutive base $s = r+2$ shows every minor with an adjacent *leading* pair is nonnegative. The mirrored identity

$$\Delta_{(r,s-1,s)}\ \Delta_{(s-2,s-1,d)} = \Delta_{(r,s-2,s-1)}\ \Delta_{(s-1,s,d)} + \Delta_{(r,s-1,d)}\ \Delta_{(s-2,s-1,s)} \tag{10}$$

handles an adjacent *trailing* pair by induction on the first row gap. Hence every order-3 minor containing at least one adjacent pair of rows is nonnegative. In both identities no sign-unknown minor occurs with a negative coefficient and no possibly-zero minor is divided out.

4.5. **Sparse rows.** Let $R = (r, u, s)$ with $r+1 < u < s$. Plücker on the five rows $r, r+1, u, s, d$ with common row u gives, with the exact orientation,

$$\Delta_{(r,u,s)}\ \Delta_{(r+1,u,d)} = \Delta_{(r,r+1,u)}\ \Delta_{(u,s,d)} + \Delta_{(r,u,d)}\ \Delta_{(r+1,u,s)}. \tag{11}$$

The divisor $\Delta_{(r+1,u,d)}$ is strictly positive in the region (8); the two d-minors are nonnegative by (7); $\Delta_{(r,r+1,u)}$ is covered by the adjacent-pair case; and $\Delta_{(r+1,u,s)}$ has first gap $u-(r+1) < u-r$. Induction on $u-r$, with the adjacent-pair case as base, proves every order-3 minor nonnegative. Together with positivity of the entries and (6), (a_n) is PF_3. □

Remark 4.1 (Duality check). For $L \geq \max(r_2, c_2)$ the substitution $R^* = (L-c_2, L-c_1, L-c_0)$, $C^* = (L-r_2, L-r_1, L-r_0)$ satisfies $\Delta_R^C = \Delta_{R^*}^{C^*}$ (reflection–transposition of the Toeplitz structure). It maps adjacent column pairs to adjacent row pairs and provides an independent cross-check of the intermediate reduction, although it is not needed once (11) is in place.

Remark 4.2. The identities (9)–(11) were verified exactly on 500 random integer matrices each, in addition to the symbolic expansion replayed in the artifact.

5. The inputs for ξ: certified range and analytic tail

We now prove Theorem 2.4. Recall the classical representation (see e.g. [8, 3])

$$\xi(\tfrac{1}{2}+z) = \int_0^\infty \Phi(u)\cosh(uz)\,du, \qquad \Phi(u) = 4\sum_{m\geq 1}\big(2\pi^2 m^4 e^{9u/2} - 3\pi m^2 e^{5u/2}\big)e^{-\pi m^2 e^{2u}}, \tag{12}$$

with $\Phi > 0$ (each summand is positive since $2\pi m^2 e^{2u}/3 > 1$). Hence

$$a_n = \frac{\gamma_n}{n!} = \frac{M_{2n}}{(2n)!}, \qquad M_t := \int_0^\infty u^t\,\Phi(u)\,du, \tag{13}$$

so $(M_t)_{t>0}$ is the moment family of a fixed positive density and in particular $a_n > 0$, proving (i).

5.1. Certified range $2 \leq n \leq 126$.

Rigorous enclosures of $\gamma_2, \ldots, \gamma_{130}$ are computed in ball arithmetic with directed outward rounding, using a discretization of (12) whose aliasing error is bounded rigorously (an earlier enclosure set lacking the alias bound was detected by internal audit and repaired; the certified set used here carries the explicit alias term). From these balls, the quantities q_n, $q_{n+1} - q_n$, and every contiguous order-3 determinant are evaluated with certified comparisons, establishing (ii)–(v) in the range $2 \leq n \leq 126$; in particular

$$q_2 = 0.4651838096951545\ldots < \tfrac{1}{2}.$$

All comparisons are fail-closed: an undecidable ball comparison aborts the run rather than reporting success.

5.2. The tilted cumulant bounds.

For the tail we study the cumulant generating function in the *shift* variable,

$$\kappa(t) := \log\int_0^\infty u^{2t}\,\Phi(u)\,du, \qquad \text{so that} \qquad \log a_n = \kappa(n) - \log(2n)!.$$

Writing $\int_0^\infty u^{2t}\Phi(u)\,du = \int_{\mathbb{R}} e^{2ts}V(s)\,ds$ with $V(s) = e^s\Phi(e^s) > 0$ (substitution $u = e^s$), the function κ is, up to the reparametrization $t \mapsto 2t$, the cumulant generating function of an exponentially tilted family; in particular $\kappa''(t) > 0$ (log-convexity) and $\kappa'''(t)$ is a positive multiple of the third cumulant of the tilt. All constants below are stated in this normalization, which is the one used by the artifact.

Proposition 5.1 (Two-sided cumulant bounds)**.** *There is an explicit threshold* $T_{71} = \big[71\pi e^{71} - \frac{9}{4}\cdot 71 - 1\big]/2 \approx 7.63\cdot 10^{32}$ *such that:*

(i) $0 < \kappa''(t) < \dfrac{4}{5t}$ *for all* $t \geq 125$*;*

(ii) $\kappa'''(t) < \dfrac{11}{20\,t^2}$ *for all* $t \geq 125$*;*

(iii) $t^2\kappa'''(t) \geq -\dfrac{3}{4}$ *for* $125 \leq t \leq T_{71}$*, and* $t^2\kappa'''(t) \geq -\dfrac{141}{142}$ *for all* $t \geq T_{71}$*. In particular* $\kappa'''(t) \geq -\frac{141}{142}\,t^{-2}$*, with constant strictly below* 1*, for every* $t \geq 125$*.*

We stress the logical roles: the *lower* bound (iii) on the third cumulant is what drives the monotonicity $q_{n+1} \geq q_n$ (it prevents the third difference of $\log a_n$ from going negative), while the upper bounds (i)–(ii) drive $q_n < 1$ and the positivity of the contiguous determinants. A one-sided upper bound on κ''' alone would be vacuous for monotonicity.

The proof of (i)–(ii) and of (iii) on the compact and banded ranges assembles 552 rigorous Arb boxes with directed endpoints, together with analytic band certificates and exact rational endpoint envelopes; the worst certified box values are

$$\text{(second moment)} \quad 1.2372479\ldots < \tfrac{5}{4}, \qquad t^2\kappa'''(t) \leq 0.1248013790\ldots < \tfrac{1}{8}$$

on the compact part, and from the exact endpoint regime onward the envelopes, e.g.

$$t\,\kappa''(t) \;\leq\; \tfrac{4660667832644329442472341693287 8}{49259028907112422032879730912332 5},$$

decrease monotonically. The bound (iii) beyond T_{71} is an *effective transfer estimate* from the dominant saddle family of the kernel to the full kernel: with $r = r(t)$ the dominant saddle parameter defined by $\pi r e^r = 2t + \frac{9}{4}r + 1$, one proves

$$\kappa'''(t) \;\geq\; -\Big[\frac{70}{r} + \frac{45450578214}{t}\Big]\frac{1}{t^2} \qquad (r \geq \tfrac{5}{2}),$$

whose bracket is $\leq \frac{141}{142}$ once $r \geq 71$, i.e. $t \geq T_{71}$; the intermediate range $125 \leq t \leq T_{71}$ is covered by the certified boxes and the analytic band certificates (with the stronger constant $\frac{3}{4}$). Full details, including the exact rational constants and their monotonicity certificates, are in the artifact.

5.3. **From cumulant bounds to the discrete conditions.** Since $\log a_n = \kappa(n) - \log(2n)!$, second and third finite differences of $\log a_n$ are, by the exact integral representations

$$\Delta^2\kappa(m) = \iint_{[0,1]^2} \kappa''\big(m+x+y\big)\,dx\,dy, \qquad \Delta^3\kappa(m) = \iiint_{[0,1]^3} \kappa'''\big(m+x+y+z\big)\,dx\,dy\,dz,$$

controlled two-sidedly by Proposition 5.1 (all arguments lie in $[n-2, n+1] \subset [125, \infty)$ for $n \geq 127$), while the factorial parts are explicit rationals. This yields, for $n \geq 127$, explicit *rational* upper envelopes $q_n \leq X_n < 1$ and $q_{n+1}/q_n \leq S_n$ (Appendix A), and with $Y_n := S_n X_n$ ($\geq q_{n+1}$) the inequalities

$$X_n < 1, \qquad Y_n < 1, \qquad H(X_n, Y_n) := 1 - Y_n - Y_n^2(1 - X_n) > 0$$

are proved by exhibiting coefficient-positive polynomials in $m = n - 127 \geq 0$. The first gives strict PF_2 in the tail. The third enters through an exact factorization of the contiguous determinant together with a *coupled monotonicity* lemma which is what makes the substitution of upper envelopes into H legitimate (Appendix A, Lemmas A.3–A.5); combined with q-monotonicity ($q_{n+2} \geq q_{n+1}$) this gives $D_n > 0$ for all $n \geq 127$.

The monotonicity $q_{n+1} \geq q_n$ in the tail is precisely where the *lower* bound of Proposition 5.1(iii) enters. Indeed

$$\begin{aligned} \log q_{n+1} - \log q_n &= \Delta^3\kappa(n-2) - B_n, \\ B_n &= \log \frac{(2n+2)(2n+1)(2n-2)(2n-3)}{\big((2n)(2n-1)\big)^2} \\ &= -\frac{2}{(n-2)^2} + O(n^{-3}). \end{aligned}$$

with B_n the logarithm of an explicit rational, and the cube representation together with (iii) bounds the cumulant term below by $-\frac{141}{142}\,(n-2)^{-2}$ (with the stronger constant $\frac{3}{4}$ below T_{71}), while $-B_n = 2(n-2)^{-2} + O(n^{-3})$: the margin is essentially a factor of two, and the exact rational bookkeeping shows $\log q_{n+1} - \log q_n > 0$ for every $n \geq 127$. An upper bound on κ''' alone would say nothing here. At the seam, the certified enclosures extend through γ_{130} and certify $q_n < 1$ up to $n = 128$, so for the ratio bounds the certified and analytic ranges genuinely overlap at $n \in \{127, 128\}$ and agree there; for the contiguous determinants the certified range $2 \leq n \leq 126$ and the analytic range $n \geq 127$ abut. Starts $m = 0, 1$ of the contiguous determinants are handled explicitly ($m = 0$ is triangular; $m = 1$ follows from $q_2 < \frac{1}{2}$ and $q_3 > 0$).

This proves (ii)–(v) for all n, completing Theorem 2.4, and with it Theorem 1.1. □

6. Sharpness

The following two propositions were verified in exact rational arithmetic; they are reproduced by a short independent script in the artifact, and were re-verified independently by the independent AI auditor layer.

Proposition 6.1. *Define $(a_n)_{0\leq n\leq 8}$ by $a_0 = 1$, $R_1 = 1$, $R_n = R_{n-1}q_n$ and*

$$(q_2, \dots, q_8) = \left(\tfrac{9}{32}, \tfrac{19}{32}, \tfrac{3}{4}, \tfrac{25}{32}, \tfrac{51}{64}, \tfrac{15}{16}, \tfrac{63}{64}\right).$$

Then q is nondecreasing with $q_2 \leq \frac{1}{2}$, and every *order-3 minor of $T(a)$ with rows and columns in $\{0, \dots, 8\}$ is nonnegative; yet*

$$\det\bigl(a_{c_j - r_i}\bigr)_{\substack{r\in(0,1,2,3)\\ c\in(2,4,5,6)}} = -\frac{809511725379305979099}{19807040628566084398385987584} < 0.$$

Hence window-PF_3 together with all ratio conditions does not imply PF_4.

Proposition 6.2. *Define $q_2 = \frac{1}{2}$ and $q_n = \frac{2n-3}{2n}$ for $n \geq 3$, with $a_0 = 1$, $R_1 = 1$. The resulting positive sequence has entire generating function, satisfies all ratio conditions (R1)–(R4), and yet its degree-3 Jensen polynomial $J^{3,0}$ has discriminant $-\frac{27}{16} < 0$ (so is not hyperbolic) and the order-4 minor with rows $(0,1,2,3)$ and columns $(1,2,3,4)$ equals $-\frac{5}{256} < 0$. The simpler sequence $b_n = 2^{-n(n-1)/2}$ ($q_n \equiv \frac{1}{2}$) likewise satisfies (R1)–(R4) and has an order-4 minor equal to $-\frac{1}{64}$.*

Corollary 6.3. *No consequence of the ratio conditions (R1)–(R4) — nor of window-verified PF_3 itself — can prove PF_4, nor degree-3 Jensen hyperbolicity, for an abstract sequence. Any extension of Theorem 1.1 up the ladder must use structural information about ξ beyond these invariants.*

Remark 6.4. This corollary sharpens, at the level of exact finite certificates, the general expectation that generic positivity hypotheses cannot climb the Pólya frequency ladder. It is also consistent with the observation of Farmer [16] that equivalences of RH tend to concentrate, rather than dilute, the original difficulty. For kernel — rather than sequence — total positivity in this landscape, see [15], where the de Bruijn–Newman kernel is certified not to be PF_5 although its low orders hold.

7. Computational methodology and reproducibility

All computations enter the proof only through certified interval (ball) arithmetic with outward rounding, performed with Arb [18] through `python-flint`. The assembly layer enforces three disciplines:

(1) *Fail-closed manifests.* Each proof stage is driven by a checker that recomputes its claims and refuses to emit its manifest unless every comparison is certified; undecidable comparisons abort.
(2) *Pinned dependencies.* A stage consumes earlier stages only through files pinned by SHA-256 hash; the final assembly re-verifies the full hash chain, so no stale or edited intermediate can enter silently.
(3) *Deterministic replay.* Every stage exposes a single command that replays its verification from the pinned inputs; the top-level replay reproduces the complete proof of Theorem 2.4 and the symbolic expansions behind Theorems 2.2 and 2.3.

The artifact further contains: the exact rational countermodels of §6 with an independent 30-line verifier; the 500 random exact tests of each Plücker identity; and the audit note documenting the detection and repair of the aliasing gap in an earlier enclosure set, which we include deliberately as part of the verification history.

Artifact. The frozen supplementary archive `xi-pf3-artifact-v1.zip` contains the proof engines, certificates, pinned SHA-256 manifest, and a one-command fail-closed replay. It is publicly available at `https://github.com/wulf758/riemann-xi-pf3` and is frozen at commit `18aa2da35cd0f45f38041365080d76a8b32ad052`. The SHA-256 digest of the deposited archive is `5D256A2580126EF4872668EBF513B431CD9861D5461786FD75F2A4D4ED9D96C0`.

8. Discussion

Theorem 1.1 places the first unconditional rung on the Toeplitz side of the Pólya frequency ladder for ξ beyond log-concavity, forty years after [3] and fifteen after the degree-3 Jensen statement of [4]. We stress what it does *not* do: by §6, the method is exactly calibrated at order 3, and no statement about PF_4, higher Jensen degrees, or RH follows. The countermodels quantify the obstruction: the ratio invariants (R_n, q_n), and even full window positivity at order 3, carry no information about order 4.

What would be needed next is a preservation mechanism special to ξ. One concrete formulation, which we leave as a problem, arises from the exact Laguerre representation of the Jensen polynomials of ξ as positive mixtures of hyperbolic Laguerre polynomials against the de Bruijn–Newman density: generic positive mixing does not preserve hyperbolicity (a two-atom counterexample exists already in degree 2), so any proof must identify a rigidity of this specific density (for the de Bruijn–Newman constant itself, see [13]) — while [15] shows that kernel-level total positivity of the same density fails at order 5 and therefore cannot be that rigidity. The difficulty of RH, viewed from this ladder, is thereby localized rather than diminished.

9. Disclosure and acknowledgements

The mathematics in this paper was produced in an extended research programme carried out by autonomous large-language-model agents operating over a shared persistent memory, under human direction, with an independent AI auditor layer; every mathematical claim was subsequently reduced to either a human-readable proof (as presented above), an exact rational certificate, or a certified ball-arithmetic computation with deterministic replay. The human author directed the programme, made all publication decisions, and takes full responsibility for the correctness of the content. The complete research log (some 1,300 hypothesis records, including the failed routes and the exact countermodels that redirected the programme) is preserved and available on request.

Appendix A. Explicit tail envelopes

We work in the normalization of §5: $\kappa(t) = \log \int_0^\infty u^{2t}\Phi(u)\,du$ and $\log a_n = \kappa(n) - \log(2n)!$. Throughout $n \ge 127$, so every cumulant argument lies in $[n-2,\, n+1] \subset [125, \infty)$. The only computer-dependent ingredients of the tail are Proposition 5.1 itself and the single certificate of Lemma A.5; Lemmas A.1–A.4 are self-contained.

Envelopes. Set

$$f_n := \frac{(n-1)(2n-3)}{n(2n-1)}, \qquad u_n := \frac{4}{5(n-2)}, \qquad v_n := \frac{11}{20(n-2)^2},$$

so that $\log q_n = \Delta^2\kappa(n-2) + \log f_n$ and $\log(q_{n+1}/q_n) = \Delta^3\kappa(n-2) + \log(f_{n+1}/f_n)$. Proposition 5.1(i)–(ii) give $\Delta^2\kappa(n-2) \le u_n$ and $\Delta^3\kappa(n-2) \le v_n$, and with $e^w \le 1/(1-w)$ for $0 \le w < 1$ we obtain the *rational* envelopes

$$q_n \;\le\; X_n := \frac{f_n}{1-u_n}, \qquad \frac{q_{n+1}}{q_n} \;\le\; S_n := \frac{f_{n+1}/f_n}{1-v_n}, \qquad q_{n+1} \;\le\; Y_n := S_n X_n.$$

Lemma A.1. *$X_n < 1$ for all $n \ge 4$.*

Proof. $X_n < 1$ is $u_n < 1 - f_n = \frac{8n-6}{2n(2n-1)}$, i.e. $4 \cdot 2n(2n-1) < 5(n-2)(8n-6)$, i.e. $24n^2 - 102n + 60 > 0$, which holds for $n \ge 4$. □

Lemma A.2. *$Y_n < 1$ for all $n \ge 6$.*

Proof. Since $Y_n = f_{n+1}/\big((1-u_n)(1-v_n)\big)$ and $(1-u_n)(1-v_n) > 1 - u_n - v_n$, it suffices that $u_n + v_n \le 1 - f_{n+1} = \frac{8n+2}{(2n+2)(2n+1)}$. Clearing the positive denominators, this is $(16n-21)(4n^2 + 6n + 2) \le 20(8n+2)(n-2)^2$, i.e. $96n^3 - 612n^2 + 574n + 202 \ge 0$, which holds for $n \ge 6$. □

Exact determinant factorization.

Lemma A.3. *With $x = q_n$, $y = q_{n+1}$, $z = q_{n+2}$,*

$$\frac{D_n}{a_n^3} = (1-y)^2 - y^2(1-x)(1-z) = (1-y)\,H(x,y) + y^2(1-x)(z-y), \qquad H(x,y) := 1 - y - y^2(1-x).$$

In particular, by q-monotonicity $z \ge y$,

$$\frac{D_n}{a_n^3} \;\ge\; (1-y)\,H(x,y).$$

Proof. Expressing the nine entries of D_n through a_n and the ratios and expanding gives $D_n/a_n^3 = 1 - 2y + xy^2 + (1-x)y^2 z$; both displayed forms are elementary rearrangements. (The identity is also replayed symbolically in the artifact and was checked on random rational sequences.) □

Coupled monotonicity. The function H is *increasing* in x, so substituting the upper envelope X_n for x directly in H would be illegitimate. The correct substitution couples the two variables through the ratio.

Lemma A.4. *Let $\widetilde{H}(x,s) := H(x,sx) = 1 - sx - s^2x^2(1-x)$. Then*

$$\frac{\partial \widetilde{H}}{\partial s} = -x\big(1 + 2sx(1-x)\big) < 0, \qquad \frac{\partial \widetilde{H}}{\partial x} = s\big(-1 + sx(3x-2)\big),$$

and since $3x-2 \le x$ for $x \le 1$, one has $sx(3x-2) \le sx^2 < 1$ whenever $sx^2 \le S_n X_n^2 \le Y_n < 1$; hence both partial derivatives are negative there. Consequently, if $q_n \le X_n$ and $q_{n+1}/q_n \le S_n$ with $Y_n = S_n X_n < 1$, then

$$H(q_n, q_{n+1}) = \widetilde{H}\big(q_n, \tfrac{q_{n+1}}{q_n}\big) \;\ge\; \widetilde{H}(X_n, S_n) = H(X_n, Y_n).$$

Proof. The two derivative identities are direct computations (the first is replayed symbolically in the artifact). Monotone descent in s then in x gives the inequality; all intermediate points satisfy the domain condition $sx^2 < 1$ because $x \le X_n$, $s \le S_n$ and $S_n X_n^2 \le Y_n < 1$ by Lemma A.2. ☐

Lemma A.5. *$H(X_n, Y_n) > 0$ for every $n \ge 127$.*

$H(X_n, Y_n)$ is an explicit rational function. More precisely, with $m = n - 127$,

$$H(X_n, Y_n) = \frac{P(m)}{(n+1)^2(2n+1)^2(5n-14)^3(20n^2-80n+69)^2},$$

where

$$\begin{aligned} P(m) = {} & 66000m^9 + 71861600m^8 + 34728981600m^7 + 9776538943040m^6 \\ & + 1766546204164425m^5 + 212447203445671919m^4 \\ & + 17002045136190509679m^3 + 872992942774568354661m^2 \\ & + 26091652886219863549980m + 345765753375849332067600. \end{aligned}$$

Every coefficient of P is strictly positive and every denominator factor is positive for $n \ge 127$, which proves the lemma directly. The identity is independently replayed in the artifact. Sample exact values are $H(X_{127}, Y_{127}) = 1.3878\ldots \times 10^{-5}$ and $H(X_{200}, Y_{200}) = 6.566\ldots \times 10^{-6}$ (the true size is $\asymp n^{-2}$).

Combining Lemmas A.2–A.5, for $n \ge 127$ we have

$$\frac{D_n}{a_n^3} \ge (1 - q_{n+1}) H(q_n, q_{n+1}) \ge (1 - Y_n) H(X_n, Y_n) > 0.$$

This is the tail closure used in §5.

Remark A.6. The certificate of Lemma A.5 genuinely lives at second order: $1 - Y_n$ and $Y_n^2(1 - X_n)$ agree at order n^{-1}, and $H(X_n, Y_n) \asymp n^{-2}$. Two nearby traps are worth recording. First, the envelope *constants* matter: doubling the exponent budgets (as happens if one miscounts the step normalization of κ) makes $H(X_n, Y_n)$ *negative* for all large n (already -1.77×10^{-4} at $n = 127$), so no certificate can exist. Second, the *coupling* of Lemma A.4 is essential: H is increasing in its first argument, and an uncoupled substitution of upper envelopes is invalid. An earlier draft of this appendix fell into both traps at once; we record them explicitly since either alone silently destroys the tail closure.

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